

UNIT - I

INTRODUCTION

Computer data often travels from one computer to another, leaving the safety of its protected physical surroundings. Once the data is out of hand, people with bad intention could modify or forge your data, either for amusement or for their own benefit.

Cryptography can reformat and transform our data, making it safer on its trip between computers. The technology is based on the essentials of secret codes, augmented by modern mathematics that protects our data in powerful ways.

- **Computer Security** - generic name for the collection of tools designed to protect data and to thwart hackers
- **Network Security** - measures to protect data during their transmission
- **Internet Security** - measures to protect data during their transmission over a collection of interconnected networks

Basic Concepts

Cryptography The art or science encompassing the principles and methods of transforming an intelligible message into one that is unintelligible, and then retransforming that message back to its original form

Plaintext The original intelligible message

Cipher text The transformed message

Cipher An algorithm for transforming an intelligible message into one that is unintelligible by transposition and/or substitution methods

Key Some critical information used by the cipher, known only to the sender & receiver

Encipher (encode) The process of converting plaintext to cipher text using a cipher and a key

Decipher (decode) the process of converting cipher text back into plaintext using a cipher and a key

Cryptanalysis The study of principles and methods of transforming an unintelligible message back into an intelligible message *without* knowledge of the key. Also called **code breaking**

Cryptology Both cryptography and cryptanalysis

Code An algorithm for transforming an intelligible message into an unintelligible one using a code-book

Cryptography

Cryptographic systems are generally classified along 3 independent dimensions:

Type of operations used for transforming plain text to cipher text

All the encryption algorithms are based on two general principles: **substitution**, in which each element in the plaintext is mapped into another element, and **transposition**, in which elements in the plaintext are rearranged.

The number of keys used

If the sender and receiver uses same key then it is said to be **symmetric key (or) single key (or) conventional encryption**.

If the sender and receiver use different keys then it is said to be **public key encryption**.

The way in which the plain text is processed

A **block cipher** processes the input and block of elements at a time, producing output block for each input block.

A **stream cipher** processes the input elements continuously, producing output element one at a time, as it goes along.

Legal, Ethical and Professional Aspects of Security

Law and Ethics in Information Security

Laws are rules that mandate or prohibit certain behavior in society; they are drawn from **ethics**, which define socially acceptable behaviors. The key difference between laws and ethics is that laws carry the sanctions of a governing authority and ethics do not. Ethics in turn are based on **Cultural mores**.

Ethical Concepts in Information Security

Cultural Differences in Ethical Concepts

- Differences in cultures cause problems in determining what is ethical and what is not ethical
- Studies of ethical sensitivity to computer use reveal different nationalities have different perspectives
- Difficulties arise when one nationality's ethical behavior contradicts that of another national group

Ethics and Education

Employees must be trained and kept aware of a number of topics related to information security, not the least of which is the expected behaviors of an ethical employee

This is especially important in areas of information security, as many employees may not have the formal technical training to understand that their behavior is unethical or even illegal

Proper ethical and legal training is vital to creating an informed, well prepared, and low-risk system user

Deterrence to Unethical and Illegal Behavior

- ü Deterrence - preventing an illegal or unethical activity
- ü Laws, policies, and technical controls are all examples of deterrents
- ü Laws and policies only deter if three conditions are present:
 - Fear of penalty
 - Probability of being caught
 - Probability of penalty being administered

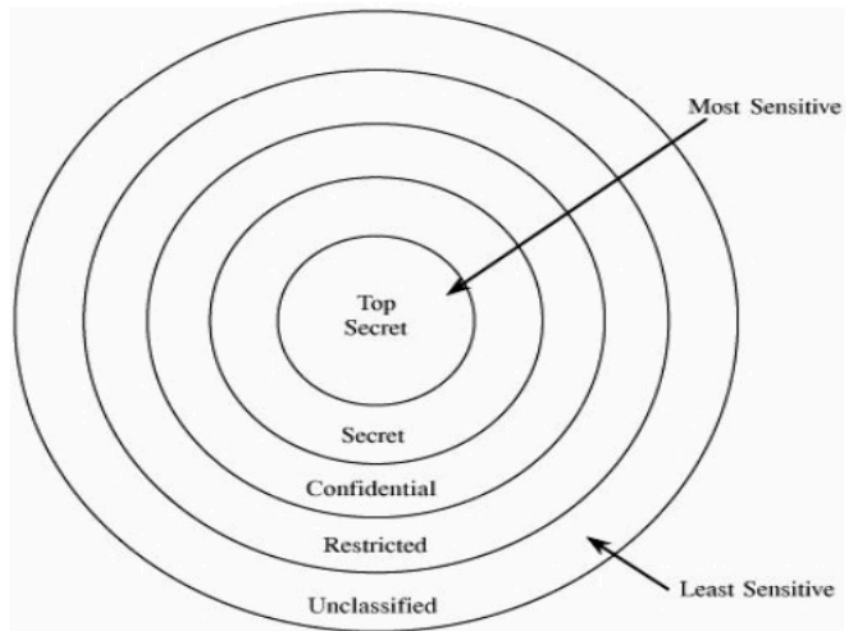
SECURITY POLICIES

To know that an operating system maintains the security we expect, we must be able to state its security policy. A **security policy** is a statement of the security we expect the system to enforce. An operating system (or any other piece of a trusted system) can be trusted only in relation to its security policy; that is, to the security needs the system is expected to satisfy.

We begin our study of security policy by examining military security policy because it has been the basis of much trusted operating system development and is fairly easy to state precisely. Then, we move to security policies that commercial establishments might adopt.

Military Security Policy

Military security policy is based on protecting classified information. Each piece of information is ranked at a particular sensitivity level, such as unclassified, restricted, confidential, secret, or top secret. The ranks or levels form a hierarchy, and they reflect an increasing order of sensitivity, as shown in Figure. That is, the information at a given level is more sensitive than the information in the level below it and less sensitive than in the level above it. For example, restricted information is more sensitive than unclassified but less sensitive than confidential. We can denote the sensitivity of an object O by rank_O . In the rest of this chapter we assume these five sensitivity levels.



Information access is limited by the **need-to-know** rule: Access to sensitive data is allowed only to subjects who need to know those data to perform their jobs. Each piece of classified information may be associated with one or more projects, called **compartments**, describing the subject matter of the information. For example, the alpha project may use secret information, as may the beta project, but staff on alpha do not need access to the information on beta. In other words, both projects use secret information, but each is restricted to only the secret information needed for its particular project. In this way, compartments help enforce need-to-know restrictions so that people obtain access only to information that is relevant to their jobs. A compartment may include information at only one sensitivity level, or it may cover information at several sensitivity levels. The relationship between compartments and sensitivity levels is shown in Figure.

